

vision**2**learn

Cyber Security Level 3

Hints
and
Tips

Unit 3



Contents

Section 1 – Understand the common types of testing in cyber security	4
Section 2 – Be able to reduce or remove potential cyber security vulnerabilities	6
Section 3 - Understand controls in cyber security	8
Section 4 - Be able to apply a cyber security control	9

Hints and Tips Unit 3

The information in this guide will be helpful to you when you are ready to complete your assessment:

Unit 3: Cyber Security Testing Vulnerabilities And Controls Level 3

Section 1 – Understand the common types of testing in cyber security

Q1. Explain what is meant by the following cyber security testing terms:

- Penetration
- Vulnerability
- Social Engineering

You will need to write an explanation for each term in the table. To assist you in providing enough information for the explanation you could include an example. Think about when each one might be used and the method involved.

Q2. Identify why cyber security testing is important

As this is an identify question, you may list or bullet point the reasons for why cyber security testing is important.

Q3. Compare types of cyber security testing by providing the pros and cons for each type in the box below:

- Social engineering testing
- Red team testing
- Vulnerability scanning
- Penetration testing

Complete the table by providing detailed pros and cons of each testing method. Think about why one method is better/not better in certain situations compared to others. Write your answers as full paragraphs and not bullet points.

Q4. Consider the following mitigations and **describe** when and how each one might be used:

Describe the different sources that cyber security professionals might use to find new threats.

- Escalation
- Patching
- User access control
- High availability
- Staff training and awareness

(continued on the next page)

Each factor in the table will need careful consideration and a detailed answer to fully describe how and when it might be used. What types of attacks are each one good for? Watch out for the key point for patching and ensure that you include this information in your answer.

Q5. In the box below, write a report to **explain**:

- Why it's important to retest following any changes made
- How the outcomes of cyber security testing can be reported
- The importance of reporting the outcomes of cyber security testing

(you may attach your report as a separate document if you prefer)

You may choose how you wish to present your report, either in the box provided or on a separate document. Ensure that you cover each of the bullet points in the question (these can be separate headings in your report if you want) and write an explanation for each one for why it's important to carry out.

Section 2 – Be able to reduce or remove potential cyber security vulnerabilities

Q1. In the table below, **Identify** vulnerabilities for the following three areas:

- Networks
- Systems
- Cloud

You may list or bullet point the vulnerabilities for each factor in the table.

Q2. This part of your assessment is within the course content on page 2, session 2.

Download the PowerPoint document – Unit 3 Section 2 Assessment – Flow Chart from the 'Time for assessment task.

Read the case study in the course content on page 2, session 2. Remember that you are working within the IT department and answer the following to demonstrate what you will do in regards to the identified vulnerabilities.

Within the Unit 3 Section 2 Assessment – Flow Chart document:

a) Produce a flow chart to demonstrate the steps needed for both vulnerabilities identified within the case study. This will include how to prioritise the vulnerabilities. See the course content for how to produce a flow chart. There is additional information within the webinar too.

Within the PowerPoint document that you've been given, you can use the flowchart symbols by selecting – insert>shapes>flowchart. Pay attention to the correct shapes for each step.

Remember: You must include reference to organisational policies/procedures within the steps

b) Provide a description for how you will apply the correct response to **one** of the **system vulnerabilities** identified in the case study. Think about what type of testing or mitigation you will use and why. Include what you will do and in which order. Have you considered anything else and decided not to apply it? Who will need the information?

c) Finally, develop an appropriate communication to mitigate future vulnerabilities. Your communication could be in the form of a leaflet, memo, email, report or any form of communication that you believe is appropriate. You should create the document for your chosen method of communication and write the necessary text to inform recipients of the security threat and what they need to do to secure this. For example, this could be a leaflet highlighting the risk to the pupils, an email to all staff detailing the steps they should take or a report to senior management explaining what they need to do to ensure that their staff understand the risk.

Once you completed the above on the PowerPoint document, save it with your name in the title and submit it with your completed assessment.

Section 3 - Understand controls in cyber security

Q1. The course material includes [three cyber security frameworks](#). Choose one of these and answer the following:

- [Explain the framework](#) - include what it's used for and when an organisation might use it.
- [Evaluate the framework](#) – what is good about this framework, not good, when is it best to use it and are there times when a different framework will be better?

You may choose one of the cyber security frameworks that are included within the course content. For example, either National Cyber Security Centre (NCSC) 10 steps, Cyber Security or Centre for Internet Security (CIS) controls or NIST Cyber Framework. Name the framework that you've chosen and make reference to it when you explain some of its features and what they might be used for. When evaluating the framework, consider its strengths and weaknesses. Also consider the other frameworks and suggest if there are times when one of those might be better or if your chosen one has any weaknesses.

Section 4 - Be able to apply a cyber security control

Q1. Using the table below, **Identify** four cyber security controls and **explain** how to apply each one

You must include the title (name) of each cyber security control and for each provide an explanation for how to apply it. Think about when you would use each one and why it's the best control for particular risks. Information for this is found in session 4, page 1 of the course content.

Q2. This part of your assessment is within the course content on page 2, session 4.

Read the case study in the course content on page 2, session 4. Download the document Unit 3 session 4 – Cyber Security Control. Use this to:

- **Implement** a basic cyber security control. There are instructions on page 2 of the course content to help you restrict access to a file on your PC for this task. You will need to take screen shots of what you do and paste them into the document provided: Unit 3 session 4 – Cyber Security Control

Q3. This question is also within the document on page 2, session 4 of the course content, in the same document entitled Unit 3 section 4 – Security Control.

- For this question, you will need to evaluate the control that you've applied in question 2. Within your evaluation, you must **justify** why this control is right for the vulnerability in the case study. You must also **explain** why another control might not be applied in this scenario.